

Marketing Budget vs. Actual Sales: A Comprehensive Statistical and Linear Regression Report

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Abstract

This report provides a comprehensive statistical analysis and linear regression modeling of the relationship between marketing budgets and actual sales. Utilizing descriptive statistics, correlation analysis, outlier detection, distribution evaluation, and residual diagnostics, this document outlines the effectiveness and predictability of marketing expenditures on sales performance.

1 Introduction

Understanding the impact of marketing expenditure on actual sales is crucial for optimizing business investments. This study analyzes a dataset comprising 222 observations of marketing budgets (in thousands of USD) and corresponding actual sales (in millions of USD). The objective is to quantify the relationship, test statistical significance, evaluate error distribution, and construct a reliable predictive model.

2 Descriptive Statistics

To understand the central tendency and spread of the data, basic descriptive statistics were computed for both variables. Table 1 summarizes the key metrics.

Statistic	Marketing Budget (thousands)	Actual Sales (millions)
Count	222.00	222.00
Mean	257.99	16.08
Std. Deviation	78.44	4.64
Min	101.91	8.01
25% Percentile	189.59	12.45
Median (50%)	262.49	15.56
75% Percentile	327.79	19.31
Max	396.88	25.98

Table 1: Descriptive Statistics of the Dataset

3 Correlation Analysis

A correlation matrix was generated to evaluate the linear association between the marketing budget and actual sales. The Pearson correlation coefficient indicates a strong positive linear relationship.

Variables	Marketing Budget (thousands)	Actual Sales (millions)
Marketing Budget (thousands)	1.000000	0.836702
Actual Sales (millions)	0.836702	1.000000

Table 2: Correlation Matrix

The correlation coefficient of $r \approx 0.8367$ demonstrates that higher marketing budgets strongly correlate with increased actual sales.

4 Linear Regression Model

A simple linear regression model was fitted to predict actual sales based on the marketing budget. The mathematical form of the model is:

$$\text{Actual Sales} = (\text{Slope} \times \text{Marketing Budget}) + \text{Intercept} \quad (1)$$

The estimated parameters derived from the analysis are:

- **Slope / Coefficient:** 0.0495
- **Intercept:** 3.3169
- **R-squared (R^2) Value:** 0.7001

This indicates that approximately 70.01% of the variance in actual sales can be explained by the marketing budget. The regression model and fitted line are visualized in Figure 1.

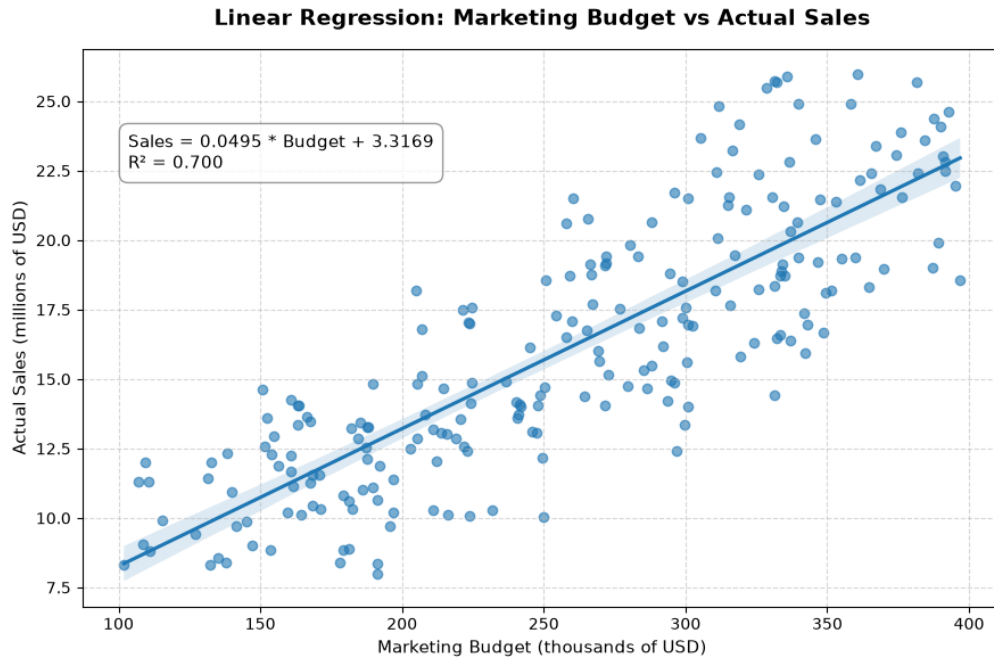


Figure 1: Linear Regression: Marketing Budget vs Actual Sales

5 Distribution Analysis

Distribution plots (histograms combined with Kernel Density Estimation curves) were generated for both variables to observe normality and spread. Figure 2 illustrates the distributions.

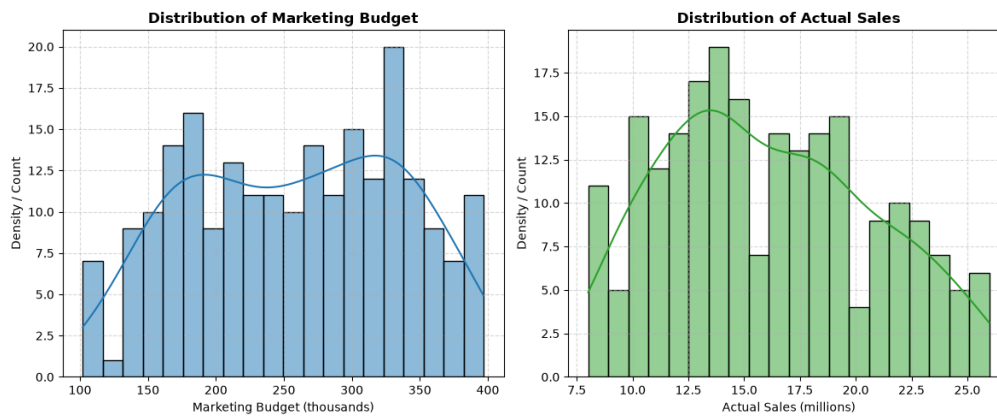


Figure 2: Distribution Plots of Marketing Budget and Actual Sales

6 Residual Diagnostics

To validate the assumptions of linear regression, a residual plot was generated by plotting the residuals (Actual minus Predicted values) against the marketing budget. Figure 3 displays the residuals.

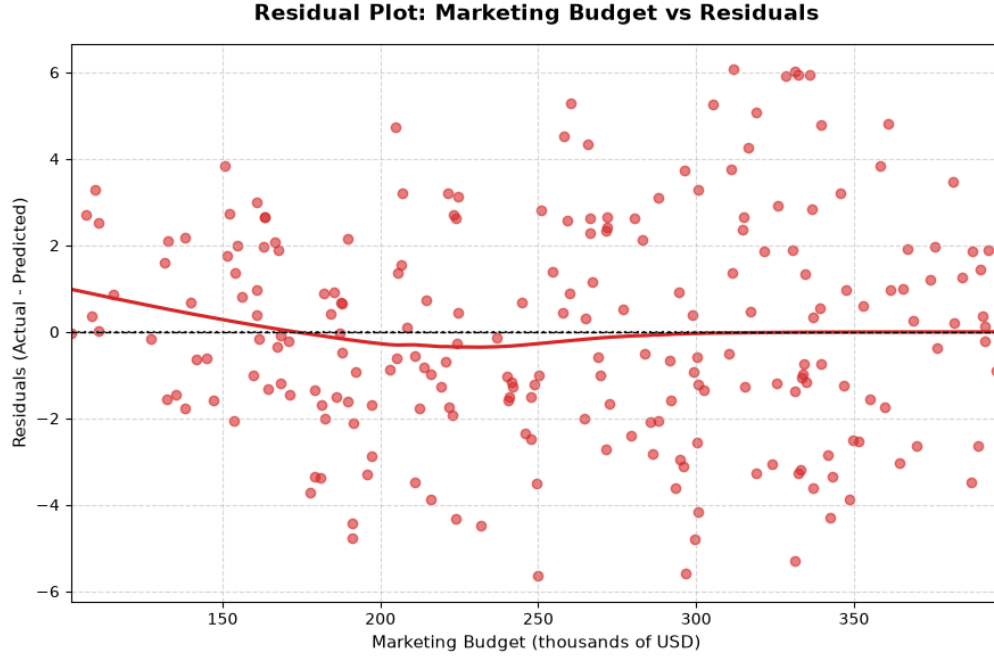


Figure 3: Residual Plot for Model Diagnostics

7 Outlier Detection

Using the Interquartile Range (IQR) method on the actual sales data, outlier analysis was performed. The analysis revealed that there are **0 outliers** in the dataset, confirming that the data points lie well within expected statistical boundaries without extreme anomalies distorting the regression line.

8 Conclusion

The analysis successfully establishes a robust linear relationship between marketing expenditures and sales performance. With a strong positive correlation (0.8367) and a high coefficient of determination ($R^2 = 0.7001$), businesses can reliably utilize the derived regression equation ($\text{Sales} = 0.0495 \times \text{Budget} + 3.3169$) to forecast future sales based on allocated marketing budgets.